
Beyond PCA: Independent Component Analysis for Audio Source Separation

A reproducible FastICA demonstration

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Project report and technical demonstration
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github.com/loukili27/ICA-Voices

Abstract

Principal Component Analysis (PCA) is widely used to reduce dimensionality by finding orthogonal directions of maximum variance. Independent Component Analysis (ICA) addresses a different question: can observed signals be expressed as mixtures of latent sources that are statistically independent? This report explains that distinction and demonstrates FastICA on three mono audio mixtures. The experiment uses 264,515 samples per observation at 44.1 kHz. FastICA converges in four iterations, produces linearly uncorrelated components, and reconstructs the observations with a relative root mean square error of approximately 4.22×10^{-16} . These values validate the numerical factorization, but they do not prove perfect source recovery because the original isolated sources are unavailable. The code, input mixtures, and separated audio outputs are provided in the companion GitHub repository.

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1 Motivation: from data compression to source separation

Multivariate observations often combine information from several latent phenomena. In audio, multiple microphones can record different linear combinations of the same speakers or instruments. In electroencephalography, each electrode measures a mixture of neural activity and artifacts. Similar mixture models arise in telecommunications, biomedical sensing, and exploratory data analysis.

PCA and ICA are both linear representation methods, but they optimize different criteria. PCA is appropriate when the objective is compression, visualization, denoising, or variance preservation. ICA is appropriate when the objective is to recover latent signals that are as statistically independent as possible [1, 3]. The classic audio illustration is the *cocktail party problem*: several sources are heard simultaneously by several microphones, and the goal is to estimate the individual sources from the recordings alone.

2 PCA and ICA: related tools, different objectives

2.1 Principal Component Analysis

For centered observations $\mathbf{x} \in \mathbb{R}^m$, PCA finds orthonormal directions that diagonalize the covariance matrix. The first principal component maximizes projected variance, the second maximizes the remaining variance subject to orthogonality, and so on [4]. If $\mathbf{U}$ contains the eigenvectors of the covariance matrix, the PCA scores are

$$\mathbf{z} = \mathbf{U}^\top(\mathbf{x} - \boldsymbol{\mu}). \quad (1)$$

The PCA scores are linearly uncorrelated. This is a second-order property determined by covariances. Uncorrelated variables are not necessarily statistically independent, except under special distributional assumptions such as joint Gaussianity.

2.2 Independent Component Analysis

ICA starts from the linear instantaneous mixing model

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t), \quad (2)$$

where $\mathbf{s}(t)$ contains unknown sources, $\mathbf{x}(t)$ contains observed mixtures, and $\mathbf{A}$ is an unknown mixing matrix. ICA estimates an unmixing matrix $\mathbf{W}$ such that

$$\mathbf{y}(t) = \mathbf{W}\mathbf{x}(t) \approx \mathbf{s}(t). \quad (3)$$

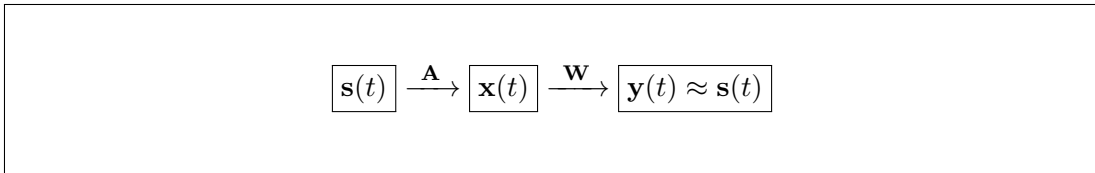


Figure 1: Linear instantaneous ICA model. FastICA estimates an unmixing transform from the observations without knowing the mixing matrix or the source waveforms.

The standard identifiable model requires mutually independent sources, a full-rank linear mixture, and at most one Gaussian source. In the determined case used here, the number of observations

equals the number of sources. ICA also has unavoidable ambiguities: recovered components may appear in any order, with arbitrary sign and scale. These ambiguities do not indicate algorithmic failure.

2.3 Why PCA alone does not solve source separation

Table 1: Conceptual comparison of PCA and ICA.

Criterion	PCA	ICA
Main objective	Maximize explained variance	Minimize statistical dependence
Output relation	Orthogonal and uncorrelated	As independent as possible
Statistics used	Primarily second-order covariance	Higher-order or information-theoretic structure
Typical use	Compression, denoising, visualization	Blind source separation, artifact removal
Component order	Ranked by explained variance	No intrinsic ranking
Ambiguities	Sign	Permutation, sign, and scale

PCA is nevertheless useful inside many ICA pipelines. Centering and PCA whitening first remove linear correlations and rescale the retained directions. ICA then rotates the whitened data to seek stronger statistical independence. In other words, PCA can be a preprocessing step for ICA, but decorrelation alone is not the final objective.

3 FastICA

FastICA uses fixed-point iterations to maximize a contrast related to non-Gaussianity [2]. The central-limit intuition is that a sum of independent variables is often more Gaussian than the original variables. Searching for non-Gaussian projections can therefore reveal the underlying sources.

For a standardized projection $y = \mathbf{w}^T \mathbf{x}$, non-Gaussianity may be approximated through negentropy. The implementation used in this project selects the smooth nonlinearity $G(y) = \log \cosh(y)$. After each update, the estimated directions are normalized and decorrelated. The parallel FastICA variant estimates all components simultaneously.

The experiment uses the **FastICA** implementation from scikit-learn with explicit unit-variance whitening [5]. This explicit setting avoids version-dependent behavior and gives every estimated source unit variance before audio export.

4 Reproducible audio demonstration

4.1 Data

The repository contains three precomputed mono WAV mixtures named `ICA mix 1.wav`, `ICA mix 2.wav`, and `ICA mix 3.wav`. Each file has the following properties:

- sampling rate: 44,100 Hz;
- resolution: 16-bit PCM;

- duration: 5.998 seconds;
- samples used: 264,515;
- number of observations: three;
- numerical rank of the observation matrix: three.

The integer samples are converted to floating point values, aligned to the shortest recording, and centered independently. The resulting matrix $\mathbf{X}$ has shape $264,515 \times 3$, with samples in rows and observed mixtures in columns.

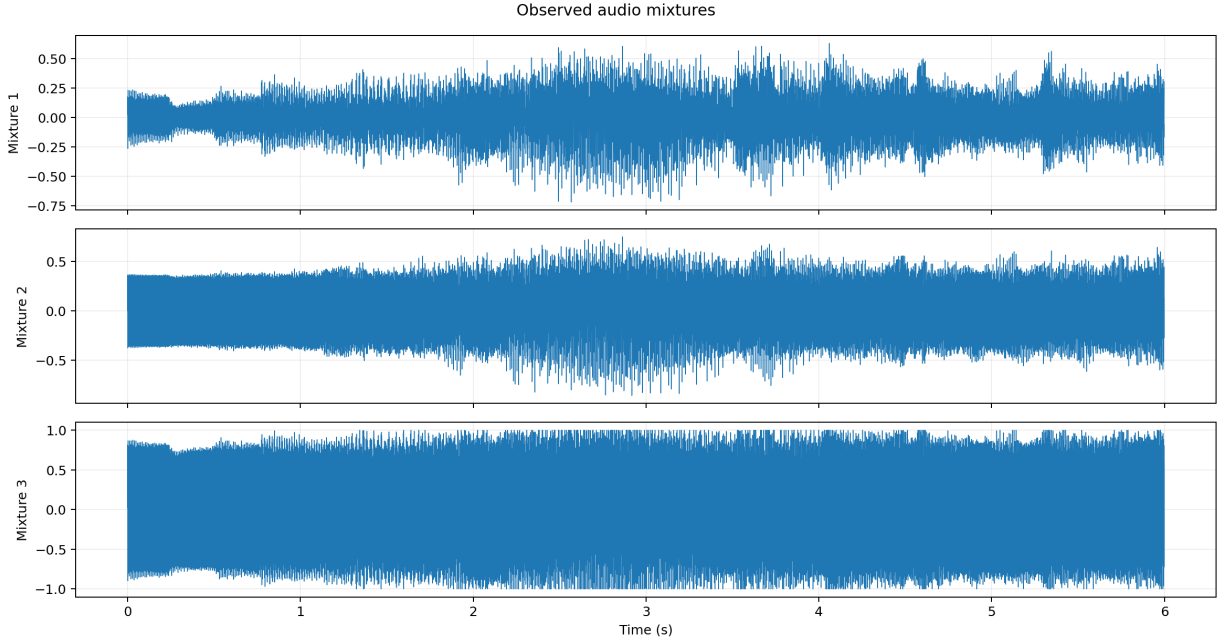


Figure 2: Waveforms of the three observed audio mixtures. Their different amplitudes and envelopes reflect different contributions from the latent sources.

4.2 Configuration

Table 2: FastICA configuration used in the reproducible notebook.

Parameter	Value
Number of components	3
Algorithm	Parallel
Whitening	Unit variance
Contrast nonlinearity	logcosh
Maximum iterations	2,000
Tolerance	10^{-5}
Random state	42

The separated components are independently normalized to 95% of the signed 16-bit range before export. This prevents clipping and integer overflow. No arbitrary volume multiplier is applied.

5 Results

5.1 Observed dependence and estimated components

The observed mixtures are strongly correlated. Their sample correlation matrix is

$$\mathbf{R}_X = \begin{bmatrix} 1.000 & 0.846 & 0.784 \\ 0.846 & 1.000 & 0.972 \\ 0.784 & 0.972 & 1.000 \end{bmatrix}. \quad (4)$$

FastICA converged in four iterations. The estimated components have an identity sample correlation matrix to the displayed precision. Their Fisher kurtosis values are $[1.401, -1.499, 2.775]$, compared with $[-1.273, 1.290, 0.931]$ for whitened PCA scores in this run. The change in higher-order statistics illustrates that ICA applies a rotation beyond PCA whitening.

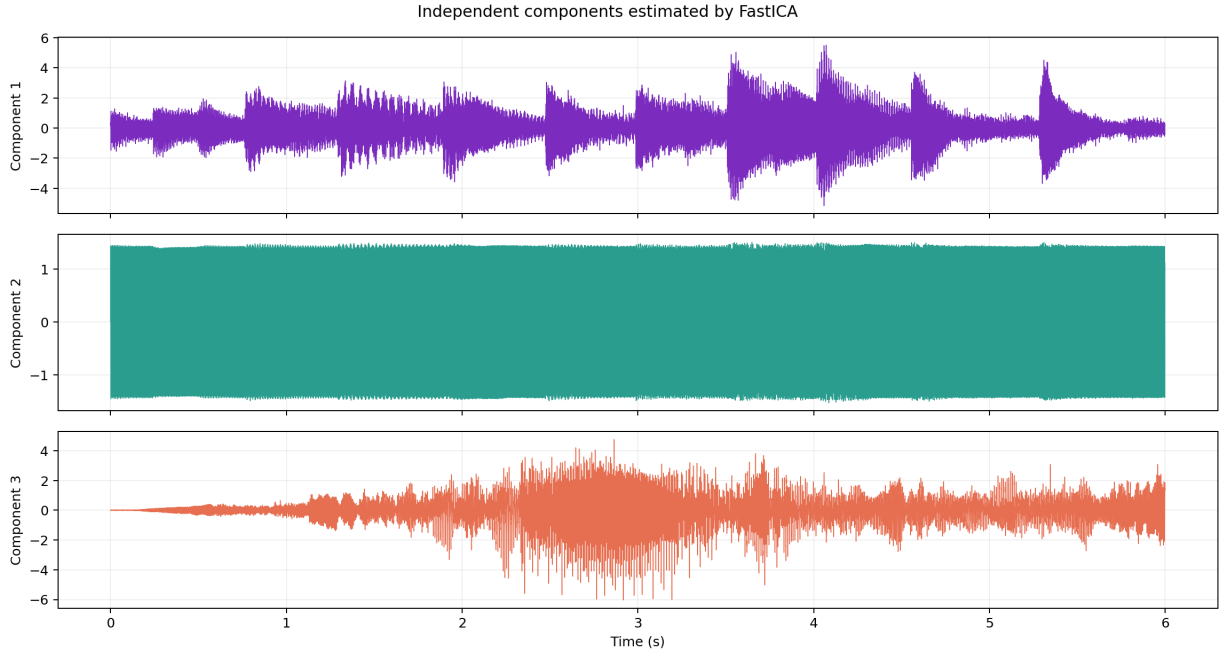


Figure 3: Waveforms of the three components estimated by FastICA. Component order, sign, and scale are intrinsically ambiguous in ICA.

5.2 Numerical checks

Table 3: Measured results from the demonstration notebook.

Metric	Result
FastICA iterations	4
Largest absolute off-diagonal ICA correlation	$< 10^{-4}$
Reconstruction RMSE	1.608×10^{-16}
Relative reconstruction RMSE	4.219×10^{-16}
Exported separated files	3

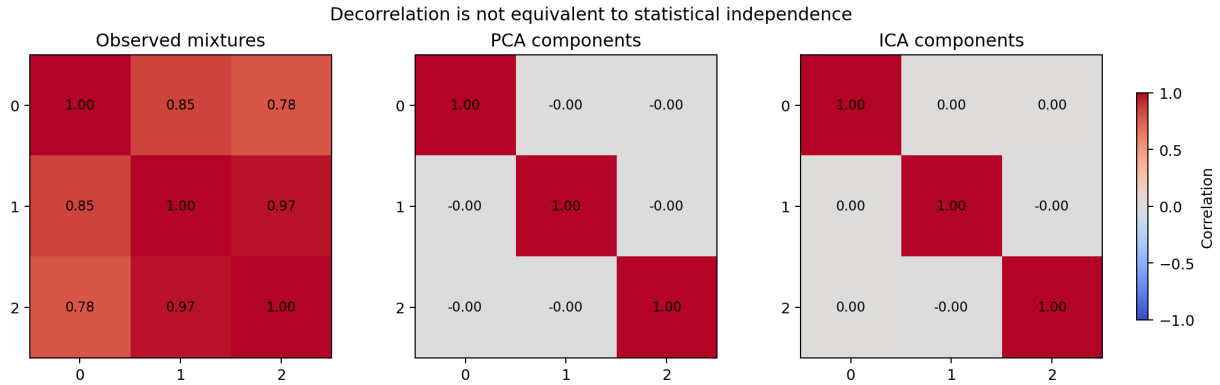


Figure 4: Correlation matrices for the observed mixtures, whitened PCA scores, and ICA estimates. PCA and ICA both remove linear correlation; this diagnostic alone therefore cannot establish statistical independence or successful source recovery.

The near-machine-precision reconstruction error confirms that the estimated mixing and unmixing transforms consistently factorize the observation matrix. It must not be interpreted as an audio source-recovery score: inverse transformation reconstructs the same mixtures on which the model was fitted.

The three exported WAV files can be inspected in `outputs/separated/` and played directly from the notebook. Listening provides a useful qualitative check, but a rigorous separation score such as SDR or SI-SDR would require access to the original isolated sources.

6 Limitations and responsible interpretation

This demonstration establishes a reproducible ICA workflow, but several limitations matter:

- The isolated reference sources and the original mixing matrix are not available. Exact recovery quality cannot therefore be quantified.
- Zero correlation is weaker than statistical independence. PCA already demonstrates this by producing uncorrelated scores.
- Instantaneous linear ICA does not explicitly model reverberation, propagation delays, non-linear mixing, or time-varying rooms.
- At least as many sufficiently distinct observations as sources are normally required in this determined formulation. Duplicating one recording does not create new information.
- ICA cannot uniquely identify multiple Gaussian sources without additional structure.

These qualifications replace the stronger claim in the draft report that the original instruments were objectively and perfectly recovered. The available evidence supports numerical convergence, decorrelation, non-Gaussian component estimation, reconstruction consistency, and qualitative listening - not a reference-based separation score.

7 Conclusion

PCA and ICA answer different questions. PCA finds an orthogonal representation that preserves variance and is ideal for dimensionality reduction. ICA seeks latent components with stronger

statistical independence and is therefore suited to blind source separation. In the audio demonstration, PCA whitening removes second-order correlations, while FastICA performs an additional rotation guided by non-Gaussianity. The notebook converges rapidly, produces three separated audio files, and reconstructs the observations to numerical precision.

The complete reproducible implementation, audio inputs, and outputs are available at

github.com/loukili27/ICA-Voices.

The report deliberately omits a code appendix: the version-controlled notebook is the authoritative implementation.

References

- [1] P. Comon, “Independent component analysis, a new concept?” *Signal Processing*, vol. 36, no. 3, pp. 287-314, 1994. DOI: [10.1016/0165-1684\(94\)90029-9](https://doi.org/10.1016/0165-1684(94)90029-9).
- [2] A. Hyvärinen, “Fast and robust fixed-point algorithms for independent component analysis,” *IEEE Transactions on Neural Networks*, vol. 10, no. 3, pp. 626-634, 1999. DOI: [10.1109/72.761722](https://doi.org/10.1109/72.761722).
- [3] A. Hyvärinen and E. Oja, “Independent component analysis: algorithms and applications,” *Neural Networks*, vol. 13, no. 4-5, pp. 411-430, 2000. DOI: [10.1016/S0893-6080\(00\)00026-5](https://doi.org/10.1016/S0893-6080(00)00026-5).
- [4] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002.
- [5] Scikit-learn developers, “FastICA API documentation,” scikit-learn.org, accessed August 24, 2026.